

Insert new Annex 180A corresponding to Clause 180 as follows:

Annex 180A

(normative)

MDIs for 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, 1.6TBASE-DR8, 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2

180A.1 Scope

This annex defines the medium dependent interface (MDI) for 200GBASE-DR1, 200GBASE-DR1-2, 400GBASE-DR2, 400GBASE-DR2-2, 800GBASE-DR4, 800GBASE-DR4-2, 1.6TBASE-DR8, and 1.6TBASE-DR8-2.

180A.2 Overview

The PMDs for 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 are specified in Clause 180. The PMDs for 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 are specified in Clause 182.

Table 180A–1 shows the number of PMDs supported by each MDI connector type.

Table 180A–1—Number of PMDs supported by each MDI connector type

MDI connector type	200GBASE-DR1, 200GBASE-DR1-2	400GBASE-DR2, 400GBASE-DR2-2	800GBASE-DR4, 800GBASE-DR4-2	1.6TBASE-DR8, 1.6TBASE-DR8-2
Single-fiber connector	1	—	—	—
Single-row 12-fiber	1, 2, or 4	1 or 2	1	—
Single-row 16-fiber	1, 2, 4, or 8	1, 2, or 4	1 or 2	1

180A.3 MDI connector requirements for multi-fiber interfaces

180A.3.1 Single-row 12-fiber interface

The MDI connector shall meet the dimensional specifications of IEC 61754-7-1 interface 7-1-9: *MPO device receptacle, angled interface*. The plug terminating the optical fiber cabling shall meet the dimensional specifications of IEC 61754-7-1 interface 7-1-1: *MPO female plug connector, down-angled interface for 2 to 12 fibres*. The MDI connector shall optically mate with the plug on the optical fiber cabling. Figure 180A–1 shows an MPO female plug connector with down-angled interface, and an MDI connector as an active device receptacle with angled interface.

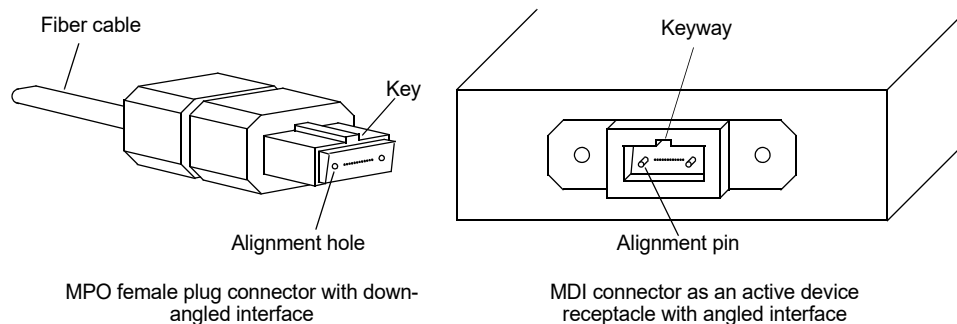


Figure 180A-1—MPO female plug with down-angled interface and MDI connector active device receptacle with angled interface

The MDI connector shall meet the interface performance specifications of IEC 61753-021-02 for performance level D/2.

180A.3.2 Single-row 16-fiber interface

The MDI connector shall optically mate with the compatible plug on the optical fiber cabling. The MDI connector shall meet the interface performance specifications of IEC 61753-021-02 for performance level D/2. 1.6TBASE-DR8 has a single-row 16-fiber interface optical lane assignment.

If the MDI connector is constructed with a connectorized fiber pigtail into an adapter, the connectorized pigtail shall meet the dimensional specifications of designation FOCIS 18 P-1x16-1-8-1-1-1 and the adapter shall meet the dimensional specifications of designation FOCIS 18 A-1-0 as defined in ANSI/TIA-604-18-A. If the MDI connector is constructed with a receptacle, it shall meet the dimensional specifications of designation FOCIS 18 R-1x16-1-8-1-1-2, as defined in ANSI/TIA-604-18-A. The plug terminating the optical fiber cabling that interconnects to either MDI connector configuration shall meet the dimensional specifications of designation FOCIS 18 P-1x16-1-8-2-1-1, as defined in ANSI/TIA-604-18-A. The MPO-16 female plug connector and MDI connector are structurally similar to those depicted in Figure 180A-1, but with 16 fibers, an offset keyway, and different pin diameters and locations.

180A.4 Optical-lane assignments

180A.4.1 Assignments for a single-row 12-fiber interface

Figure 180A-2 shows the generic layout of a single-row 12-position optical connector specified for single lane, 2-lane and 4-lane optical PMDs. The four transmit and four receive optical lanes shall occupy the positions depicted in Figure 180A-2 when looking into the MDI connector receptacle with the connector keyway feature on top. The four center positions are unused. Figure 180A-2 also provides details on the ordering of the optical connector positions.

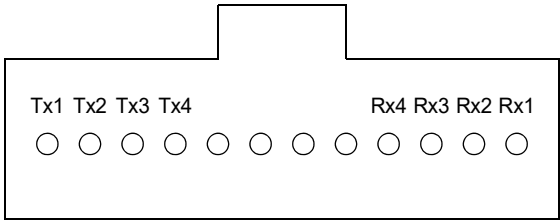


Figure 180A–2—Optical lane assignments for a single-row 12-position connector

When only a single 1-lane PMD (200GBASE-DR1 or 200GBASE-DR1-2) is implemented with this optical connector, the optical PMD lanes shall be assigned to the optical connector positions Tx1 and Rx1, as shown in Table 180A–2, regardless of whether fibers are populated in the remaining optical connector positions.

When only a single 2-lane PMD (400GBASE-DR2 or 400GBASE-DR2-2) is implemented with this optical connector, the optical PMD lanes shall be assigned to the optical connector positions Tx1, Tx2, Rx1, and Rx2, as shown in Table 180A–2, regardless of whether fibers are populated in the remaining optical connector positions.

When a single 4-lane PMD (800GBASE-DR4 or 800GBASE-DR4-2) is implemented with this optical connector, the optical PMD lanes shall be assigned to the optical connector positions Tx1 through Tx4 and Rx1 through Rx4, as shown in Table 180A–2.

Alternatively, combinations of 1-lane optical PMDs (200GBASE-DR1 or 200GBASE-DR1-2) and 2-lane optical PMDs (400GBASE-DR2 or 400GBASE-DR2-2) are permissible when fibers are populated in all of the optical connector positions. For 2-lane PMDs, the optical lanes shall be assigned to the optical connector positions for a given 400GBASE-DR2 or 400GBASE-DR2-2 PMD # in Table 180A–2. Table 180A–3 shows an example of a combination of the mapping of several optical PMDs to a single row, 12-position optical connector with fibers populated in optical connector positions Tx1 through Tx4 and Rx1 through Rx4.

NOTE—Implementations that support multiple combinations of PHY types on the same MDI connector must be configured appropriately for interoperability with the connected link partners. Selecting the appropriate configuration requires knowledge of the fiber plant and the link partners.

Table 180A–2—Mapping PMD signals to positions for 12-position optical connectors

<PMD #>:<PMD signal>			
200GBASE-DR1 ^a	400GBASE-DR2 ^a	800GBASE-DR4 ^a	Connector position
0:SL0	0:SL0	0:SL0	Tx1
1:SL0	0:SL1	0:SL1	Tx2
2:SL0	1:SL0	0:SL2	Tx3

Table 180A–2—Mapping PMD signals to positions for 12-position optical connectors

<PMD #>:<PMD signal>			
200GBASE-DR1 ^a	400GBASE-DR2 ^a	800GBASE-DR4 ^a	Connector position
3:SL0	1:SL1	0:SL3	Tx4
0:DL0	0:DL0	0:DL0	Rx1
1:DL0	0:DL1	0:DL1	Rx2
2:DL0	1:DL0	0:DL2	Rx3
3:DL0	1:DL1	0:DL3	Rx4

^a In this table, the mapping of 200GBASE-DR1, 400GBASE-DR2, and 800GBASE-DR4 is shown, however this may equally apply to 200GBASE-DR1-2, 400GBASE-DR2-2, and 800GBASE-DR4-2.

Table 180A–3—Example mapping on a 12-position optical connector

<PMD #>:<PMD signal>		
200GBASE-DR1	400GBASE-DR2	Connector position
—	0:SL0	Tx1
—	0:SL1	Tx2
2:SL0	—	Tx3
3:SL0	—	Tx4
—	0:DL0	Rx1
—	0:DL1	Rx2
2:DL0	—	Rx3
3:DL0	—	Rx4

180A.4.2 Assignments for a single-row 16-fiber interface

Figure 180A–3 shows the generic layout of a single-row 16-position optical connector specified for 1-lane, 2-lane, 4-lane and 8-lane optical PMDs. The eight transmit and eight receive optical lanes shall occupy the positions depicted in Figure 180A–3 when looking into the MDI connector receptacle with the connector keyway feature on top. Figure 180A–3 also provides details on the ordering of the optical connector positions.

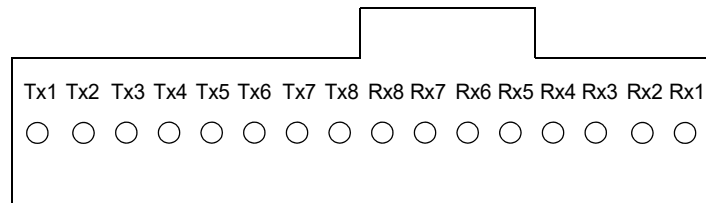


Figure 180A–3—Optical lane assignments for a single-row 16-position connector

When only a single 1-lane PMD (200GBASE-DR1 or 200GBASE-DR1-2) is implemented with this optical connector, the optical PMD lanes shall be assigned to the optical connector positions Tx1 and Rx1, as shown in Table 180A–4, regardless of whether fibers are populated in the remaining optical connector positions.

When only a single 2-lane PMD (400GBASE-DR2 or 400GBASE-DR2-2) is implemented with this optical connector, the optical PMD lanes shall be assigned to the optical connector positions Tx1, Tx2, Rx1, and Rx2, as shown in Table 180A–4, regardless of whether fibers are populated in the remaining optical connector positions.

When only a single 4-lane PMD (800GBASE-DR4 or 800GBASE-DR4-2) is implemented with this optical connector, the optical PMD lanes shall be assigned to the optical connector positions Tx1 through Tx4 and Rx1 through Rx4, as shown in Table 180A–4, regardless of whether fibers are populated in the remaining optical connector positions.

When a single 8-lane PMD (1.6TBASE-DR8 or 1.6TBASE-DR8-2) is implemented with this optical connector, the optical PMD lanes shall be assigned to the optical connector positions Tx1 through Tx8 and Rx1 through Rx8, as shown in Table 180A–4.

Alternatively, combinations of 1-lane optical PMDs (200GBASE-DR1 or 200GBASE-DR1-2) and 2-lane optical PMDs (400GBASE-DR2 or 400GBASE-DR2-2) and 4-lane optical PMDs (800GBASE-DR4 or 800GBASE-DR4-2) are permissible when fibers are populated in all of the optical connector positions. For 2-lane PMDs, the optical lanes shall be assigned to the optical connector positions for a given 400GBASE-DR2 or 400GBASE-DR2-2 PMD # in Table 180A–4. For 4-lane PMDs, the optical lanes shall be assigned to the optical connector positions for a given 800GBASE-DR4 or 800GBASE-DR4-2 PMD # in Table 180A–4. Table 180A–5 shows an example of a combination of the mapping of several optical PMDs to a single row, 16-position optical connector with fibers populated in optical connector positions Tx1 through Tx8 and Rx1 through Rx8.

NOTE—Implementations that support multiple combinations of PHY types on the same MDI connector must be configured appropriately for interoperability with the connected link partners. Selecting the appropriate configuration requires knowledge of the fiber plant and the link partners.

Table 180A–4—Mapping PMD signals to positions for 16-position optical connectors

<PMD #>:<PMD signal>				
200GBASE-DR1 ^a	400GBASE-DR2 ^a	800GBASE-DR4 ^a	1.6TBASE-DR8 ^a	Connector position
0:SL0	0:SL0	0:SL0	0:SL0	Tx1
1:SL0	0:SL1	0:SL1	0:SL1	Tx2
2:SL0	1:SL0	0:SL2	0:SL2	Tx3
3:SL0	1:SL1	0:SL3	0:SL3	Tx4
4:SL0	2:SL0	1:SL0	0:SL4	Tx5
5:SL0	2:SL1	1:SL1	0:SL5	Tx6
6:SL0	3:SL0	1:SL2	0:SL6	Tx7
7:SL0	3:SL1	1:SL3	0:SL7	Tx8
0:DL0	0:DL0	0:DL0	0:DL0	Rx1
1:DL0	0:DL1	0:DL1	0:DL1	Rx2
2:DL0	1:DL0	0:DL2	0:DL2	Rx3
3:DL0	1:DL1	0:DL3	0:DL3	Rx4
4:DL0	2:DL0	1:DL0	0:DL4	Rx5
5:DL0	2:DL1	1:DL1	0:DL5	Rx6
6:DL0	3:DL0	1:DL2	0:DL6	Rx7
7:DL0	3:DL1	1:DL3	0:DL7	Rx8

^a In this table the mapping of 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 is shown, however this may equally apply to 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2.

Table 180A–5—Example mapping on a 16-position optical connector

<PMD #>:<PMD signal>			
200GBASE-DR1	400GBASE-DR2	800GBASE-DR4	Connector position
—	0:SL0	—	Tx1
—	0:SL1	—	Tx2
2:SL0	—	—	Tx3
3:SL0	—	—	Tx4
—	—	1:SL0	Tx5
—	—	1:SL1	Tx6
—	—	1:SL2	Tx7

Table 180A–5—Example mapping on a 16-position optical connector

<PMD #>:<PMD signal>			
200GBASE-DR1	400GBASE-DR2	800GBASE-DR4	Connector position
—	—	1:SL3	Tx8
—	0:DL0	—	Rx1
—	0:DL1	—	Rx2
2:DL0	—	—	Rx3
3:DL0	—	—	Rx4
—	—	1:DL0	Rx5
—	—	1:DL1	Rx6
—	—	1:DL2	Rx7
—	—	1:DL3	Rx8